Confidence Sidecar: 5-Page Brief

1,497-segment B3 decode – the operational story

Source. Per-token confidence sidecar from the 1,497-segment B3 decode (`/tmp/vsp_b3_1497_out/confidence-172610.json`), joined with baseline ground truth (IS, WER, NEA F1). 23,261 hyp words. Hypotheses are bit-identical to the baseline decode – the 59.17% pooled vs 64.05% mean-of-segment WER gap is purely the pooled-vs-averaged divergence on a heavy-tailed distribution.
Scope of this brief. This document summarises the deployment-relevant findings of the full confidence analysis: how strongly confidence predicts quality, where to set the gate, what slips through, and the failure-mode taxonomy. *Per-word band calibration is not covered here* – see the comprehensive 17-page report for the calibration analysis (overall ECE, stratified band reliability, and the three-tier display policy).

Headline findings

TL;DR. The average per-word confidence inside a segment (mean of the model's max-softmax probabilities, one per output word) predicts the segment's overall quality with Pearson $r = +0.84$ against the Intelligibility Score. That single number is enough to filter bad outputs out at runtime. The remaining failures break into a small set of categories that need beam-aggregation or topic context to detect.

Term key. IS = Intelligibility Score (0–5, mean = 2.52). **NIV-Y** = "clearly intelligible" (IS $\geq$ 3.80). **NIV-N** = "not intelligible" (IS $<$ 2.0). **mean_word_prob** = per-segment average of per-word probs (each per-word prob = min over its sub-tokens). **WER / WWER** = word error rate / weighted WER (content tokens count 2×).

1. **Mean word probability is the single best confidence aggregate** – Pearson $r = +0.837$ with IS, -0.714 with WER, -0.749 with WWER (n = 1,427). Geomean prob, mean entropy, mean margin, and `frac_p_ge_0.85` all sit within ~ 0.03 r of this – different shadows of the same signal.
2. **Confidence is a strong segment-level filter** – AUC = 0.917 for NIV-N, 0.921 for NIV-Y from `mean_prob` alone. At `mean_word_prob` $\geq$ 0.80 we keep **78**% of NIV-Y and admit only **9** NIV-N out of 405 trusted segments (2.2% false-trust). For zero-tolerance use, add `duration` $\geq$ `1.5s AND` `mean_entropy` $\leq$ 0.7 $\rightarrow$ zero false-bads at $\sim$ 30% recall.
3. **Hallucinations mostly self-flag** – of 223 fluent-but-wrong segments, only **3** (1.3%) have `mean_prob` $\geq$ 0.85. Median `mean_prob` 0.541 vs 0.759 healthy; entropy AUC ties with prob AUC. The literature's worst-case "fluent fabrications at $p > 0.95$" failure mode is rare in this data.
4. **Different failure modes need different signals** – 503 IS<2 segments classify as Wrong Topic 68%, Hallucination 10%, Right Topic Wrong Details 15%, Signal Loss 0.2%, Accumulated Errors 7%. Hallucination + Signal Loss ($\approx$10%) are catchable from confidence + length-ratio. The remaining $\sim$90% require the reference (semantic / NEA) or a runtime substitute – **Mission 6** (all-20-beams) and **Mission 8** (topic-LM) are the next-sprint priorities.

1 What confidence is, and what it predicts

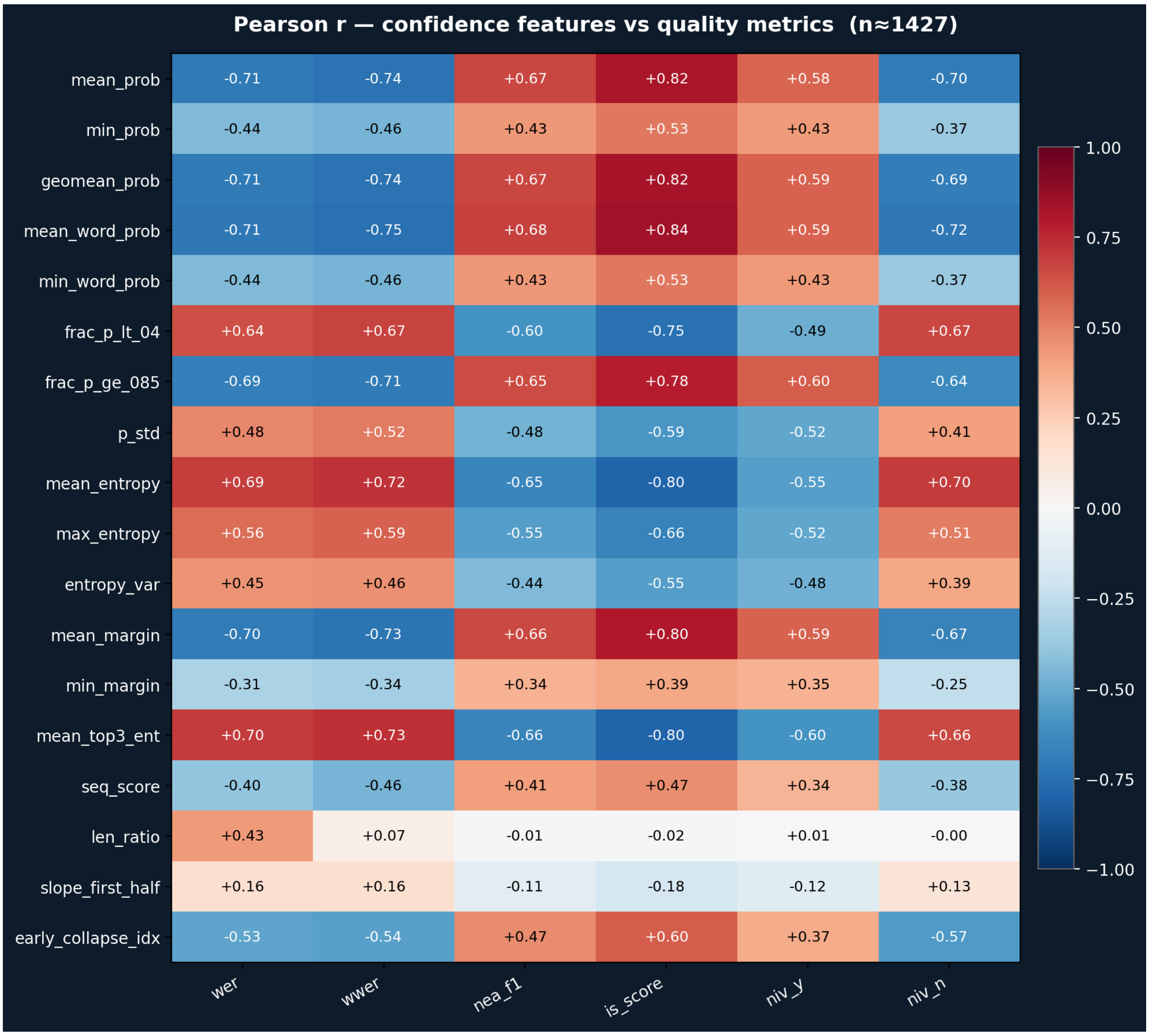

Figure 1: Pearson correlation matrix. Rows: confidence-derived features. Columns: quality metrics. Mean-aggregates of probability, entropy, margin, and "green-band-fraction" all hit $|r| \geq 0.7$ against IS / WER / WWER – they're interchangeable as quality signals. `min_word_prob` is meaningfully weaker (one bad word can occur in an otherwise-fine segment). Spearman ρ tracks Pearson r within 0.06 for confidence-vs-IS pairs, so the relationship is not a linearity artifact.

Top-5 features by $|r|$ with IS: `mean_word_prob` +0.837, `geomean_prob` +0.822, `mean_prob` +0.818, `mean_entropy` −0.804, `mean_margin` +0.803. Three of the four leading signals explain ∼ 65% of IS variance with a single linear term. Top-3-only entropy is $r = +0.946$ redundant with full-distribution entropy – the cheap version isn't enough; genuine beam-level information needs all 20 hypotheses (Mission 6).

Practical reading. For runtime quality estimation we don't need anything fancier than averaging the per-word probabilities the model already produces.

2 Filtering: operating points and what slips through

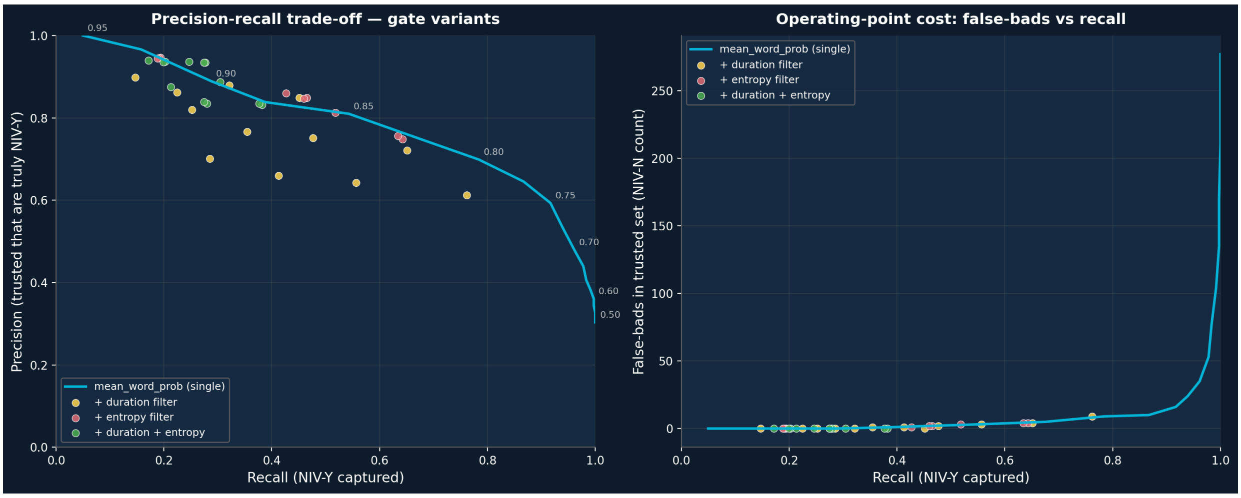

Figure 2: Left: precision-recall trade-off. Cyan curve sweeps `mean_word_prob` alone; coloured dots overlay duration / entropy / triple-signal gates. Right: false-bad count (NIV-N inside trusted set) as recall grows. Adding constraints trades recall faster than it adds precision, but enables zero-false-bad operating points for high-stakes use.

Operating point	Gate	Prec.	Recall	False-bads	Vol. %
Loose (high recall)	`mwp` ≥ 0.75	59.3%	91.7%	16	39.1%
Balanced	`mwp` ≥ 0.80	**69.9%**	**78.4%**	9	28.4%
Strict precision	`mwp` ≥ 0.85	81.0%	54.3%	3	17.0%
+ duration filter	`mwp` ≥ 0.85 `AND dur` ≥ 1.5`s`	87.9%	32.1%	0	9.3%
Zero-false-bad	`mwp` $\geq$ 0.85 `AND dur` $\geq$ `1.5s AND ent` $\leq$ 0.7	**88.7%**	**30.5%**	**0**	8.7%

2.1 Who slips through? False-good profile

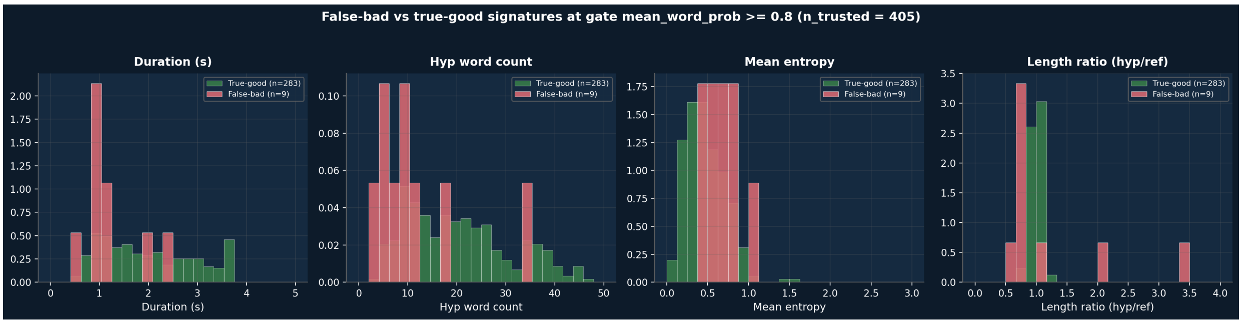

Figure 3: Density of true-good (green) vs false-bad (coral) at `mean_word_prob` ≥ 0.80. **The smoking gun: false-bads are systematically SHORT.** Median duration 0.98s vs 1.86s; median hyp word count 9 vs 19; mean entropy is +0.20 higher.

False-bads are systematically segments where the model has too little material to constrain its output – short clips with a handful of words give the language prior room to commit to a fluent-but-wrong answer. **Visual cues:** segment duration is a free runtime proxy for visual quality. A real visual quality model (lip occlusion, face-frame coverage, head pose) would catch more, but is out-of-pipeline today. **Topic context:** a surrounding-sentence LM conditioned on the source video's topic could rescore the hyp and flag drift; out-of-pipeline today. **Beam disagreement** (Mission 6) is the most tractable next-sprint signal for the truly fluent-and-plausible cases that confidence alone cannot detect.

3 Failure modes in confidence space

The 503 IS<2 segments classify into five mutually-exclusive categories (March 2026 deck taxonomy, evaluated $1 \to 5$). Each segment gets exactly one label.

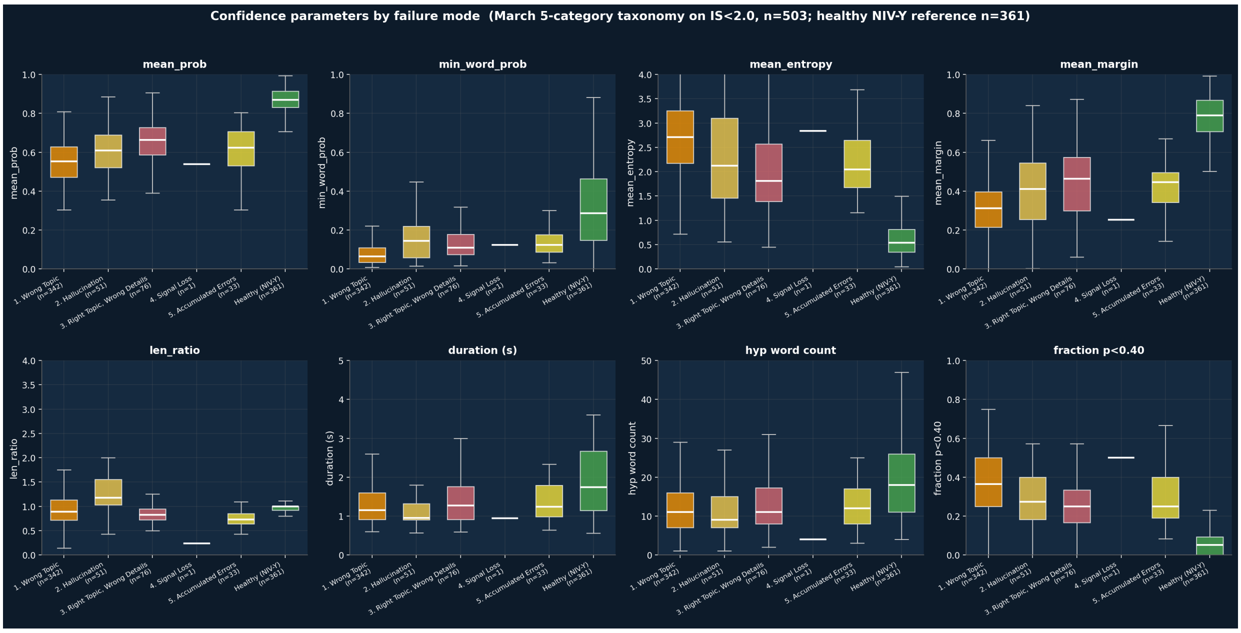

Figure 4: Boxplots of eight confidence parameters across the five March failure categories plus the healthy NIV-Y reference. Wrong Topic dominates ($\sim$ 68%) – the model commits decisively to a wrong domain. Confidence collapses uniformly on every parameter for failures *except* `mean_prob`, where Wrong Topic still hovers in the medium band – the deck's "confidence color cannot detect topic substitution" caveat.

Category	**n**	**m_prob**	**min_wp**	**m_ent**	**dur (s)**	**NEA**	**WER%**	**IS**
1. Wrong Topic	342	0.55	0.08	2.74	1.39	0.7	137	1.03
2. Hallucination	51	0.59	0.16	2.27	1.19	5.6	145	1.39
3. Right Topic, Wrong Details	76	0.65	0.16	1.97	1.42	4.4	71	1.65
4. Signal Loss	1	0.54	0.12	2.84	0.94	0	100	1.91
5. Accumulated Errors	33	0.62	0.13	2.14	1.37	31.5	51	1.77
Healthy (NIV-Y)	361	**0.86**	**0.32**	**0.62**	**1.93**	**84.6**	**22**	**4.33**

3.1 Trajectory shape predicts outcome

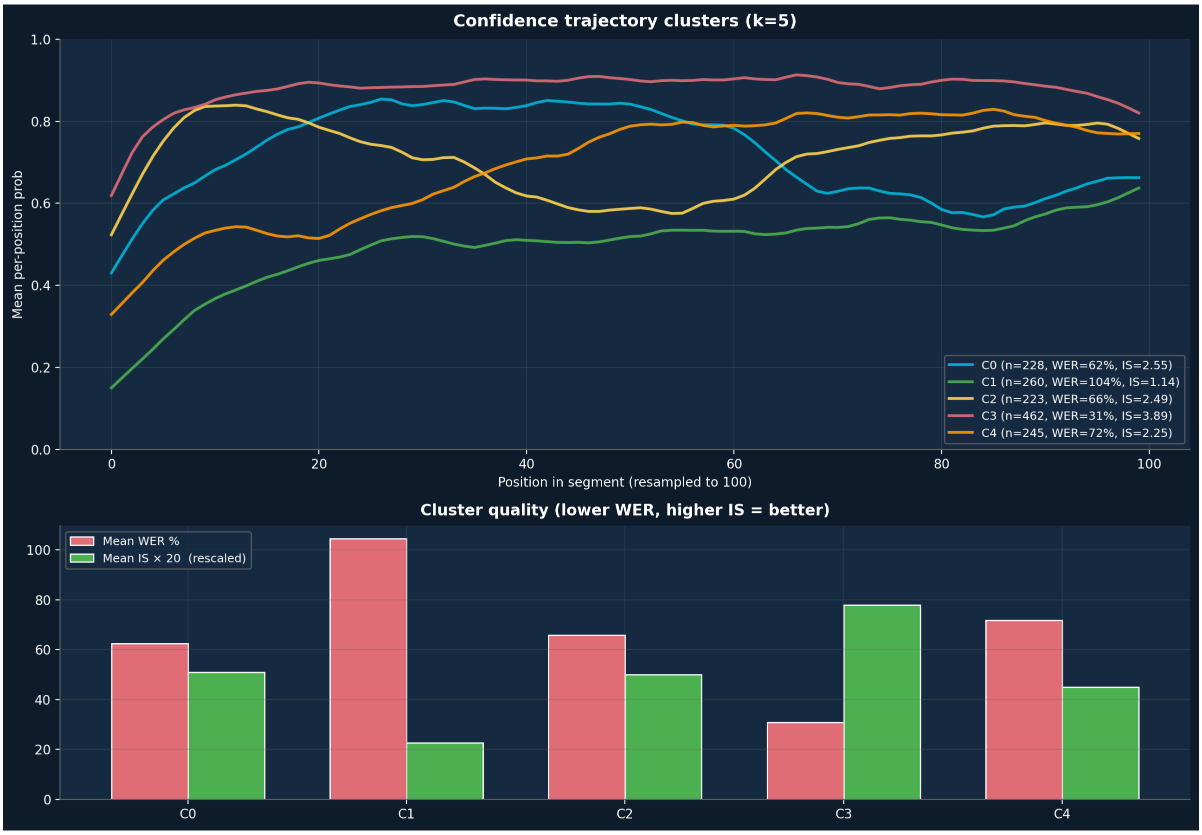

Figure 5: Top: $k = 5$ cluster centroids on length-normalised per-position confidence trajectories. Bottom: per-cluster mean WER (coral) and mean IS rescaled 20× (green). Flat-high cluster (C3): mean WER **31**%, IS 3.89, 64% NIV-Y. Uncertain-throughout (C1): mean WER **104**%, IS 1.14, **92**% NIV-N. The shape difference is visible after just a few tokens – viable as a runtime "give up early" hook (Mission 11).

4 What confidence cannot tell us, and what's next

Confidence has three blind spots, all visible in the failure-mode breakdown:

1. **Wrong Topic** (68% of bad segments) – model commits decisively to a wrong domain; `mean_prob` sits in the medium band, no confidence signal sharp enough to flag. Needs semantic similarity (reference-required) or a topic-conditioned LM scoring the hyp for surprise relative to the source video's topic. *Mission 8.*
2. **Right Topic, Wrong Details** (15%) – closest of any failure mode to healthy in confidence space. Model knows the topic but loses the specific entities. Needs NEA F1 (reference-required) or beam disagreement – if the chosen beam dropped a number that other beams kept, that's a flag. *Mission 6 (all-20-beams).*
3. **The "billion → million" problem** – single high-confidence wrong content word inside an otherwise-fine segment. Bypasses all current signals because the model is decisive on the wrong answer. Needs source-context priors or visual disambiguation. *Long-term.*

Recommendations.

1. Use `mean_word_prob` $\geq$ 0.80 as the runtime quality gate. Already exposed via `make_report.py -word-confidence` as `sentence_confidence`.
2. For per-word coloring display, gate the rendering on segment-level `mean_word_prob` $\geq$ 0.82 (the band-reliability threshold from the full report). Below that, hide or banner.
3. Mission 6 (all-20-beams capture) – highest-leverage next-sprint item. Top-3 entropy is $r = 0.95$ redundant with full entropy; we need genuine beam disagreement.
4. Mission 8 (topic-conditioned LM for hyp rescoring) – handles the Wrong Topic hole, 68% of bad segments.
5. Mission 11 (mid-decode trajectory monitoring) – runtime "give up early" hook from Section 3.

For calibration analysis (overall ECE, stratified band reliability, three-tier display policy) and the reproduce command, see the full 17-page report at `docs/confidence/confidence_full_analysis.pdf`.